Turbulence modulation via dilute surfactant additives in experimental Taylor–Green flow

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We investigate how dilute solutions of the cationic surfactant CTAC/NaSal modulate turbulence driven by inertial forcing, and reveal that this modulation cannot be represented by a single relaxation time of the solution. Instead, our results indicate that the modulation involves at least two types of viscoelastic structures with distinct time scales. To this end, we use an experimental Taylor–Green-like flow driven by rotating blades of variable speed. Varying the rotation rate controls the ratio of the surfactant relaxation time to the characteristic flow time scale, spanning a wide range of Weissenberg and Reynolds numbers. We identify two distinct regimes of modulation depending on the forcing frequency. At low rotation rates, both the mean flow and the velocity fluctuations are intensified. This intensification is driven by a resonance between the periodic forcing imposed by the blades and the formation of large-scale micellar structures, producing jet-like flows between the blades. Subtracting the periodic flow component through a triple decomposition reveals a second regime characterized by turbulence attenuation, which begins once the Weissenberg number based on the Kolmogorov time scale of the Newtonian flow exceeds unity. As the rotation rate is further increased, turbulence attenuation progressively vanishes, which we attribute to the breakdown of the micellar network under strong shear.

I. INTRODUCTION

Since the discovery that trace amounts of dissolved polymer could lead to drag reduction in pipe flows [1], the use of additives for drag reduction in wall-bounded flows has become well established. Besides polymer solutions, it has since been discovered that self-assembling surfactant solutions share similar effects [2], owing to the fact that dilute surfactant solutions can form wormlike micelles whose relaxation dynamics mimic those of polymer chains while resisting mechanical degradation.

Beyond the applications of drag reduction, it was shown that viscoelastic additives modulate turbulence more broadly, reshaping the energy spectrum and producing an elastic range governed by additive elasticity rather than inertia at the small scales [3, 4], with surfactant-specific micellar structures shown to alter vortex shedding and turbulence statistics directly [5]. The most common picture underlying most of these observations is a time scale matching argument: turbulent eddies with a turnover time shorter than the relaxation time of the additive interact appreciably with it, leading to a truncation of the energy cascade below a corresponding length scale, the Lumley scale [2, 6]. This time scale matching is conveniently expressed through the Weissenberg number,

$$Wi \equiv \frac{\tau_s}{\tau_\eta}, \tag{1}$$

defined as the ratio between the relaxation time τ_s of the viscoelastic solution and a characteristic time scale of the turbulent flow, most commonly taken as the Kolmogorov time scale τ_η of the Newtonian flow. Broadly speaking, turbulence modulation by the additive is expected to become significant once $Wi \gtrsim 1$, since only then do turbulent eddies become fast enough for the additive to respond elastically rather than viscously. A refined version of the argument predicts a somewhat smaller truncation scale on the basis of an elastic energy budget between the additive elasticity and the turbulent eddies [7].

In spite of academic and commercial interest, comparatively few studies address the behavior of dilute surfactant solutions in the context of three-dimensional bulk turbulence, with most of the literature focused on wall-bounded or quasi-two-dimensional flows. A notable exception, and one directly relevant to the present

work, is the study of Horimoto and Goto [8], who investigated the modulation of confined turbulence sustained by the precession of a spherical container using dilute solutions of both CTAC and polyethylene oxide (PEO). Using the same Lumley-scale/elastic-cascade arguments described above, they showed that small-scale turbulent eddies are drastically suppressed by the CTAC additive while a pair of large-scale coherent vortex tubes survives essentially unmodified. This result illustrates both the strength of the time scale matching argument in predicting which additive is effective in a given flow, and the fact that the interaction between additive elasticity and large-scale, coherent flow structures can play a central role in determining the overall response of the flow.

In the aforementioned study, energy is transmitted to the flow and drives the turbulence via the viscous stresses against the wall of the container which rotates in a uniform precessing motion. In contrast to this, turbulence can also be driven mainly through pressure forces, such as through the use of impellers which generate a spatially localized, time-dependent forcing. For simplicity, this type of forcing is hereinafter simply denominated inertial forcing. It was shown that turbulence changes qualitatively between inertial and viscous forcing [9], which is an indication that the modulation mechanism may also change between them. However, the nature of this change still remains an open question.

This gap is compounded by the fact that the rheology of very dilute surfactant solutions remains difficult to pin down, with reported values of viscosity and relaxation time of CTAC/NaSal solutions (such as the one considered in the present experiment) varying considerably between studies [10, 11]. In spite of the diverging measurements and reports that some of the dilute solutions behave as a Newtonian fluid under rheological measurements [10], such solutions are known to produce measurable drag reduction [12], which further reinforces the need to analyze such solutions more in depth.

Motivated by this gap, here we investigate how dilute CTAC solutions modulate inertially driven turbulence, considering the case of a four-rotor, Taylor–Green-like experiment across a wide range of Reynolds and Weissenberg numbers. The paper is organized as follows. Section II describes the experimental apparatus, velocity measurement and data processing methods, as well as the surfactant solutions. Subsequently, section III characterizes the Newtonian turbulence with estimates of the Kolmogorov scales and corresponding Weissenberg numbers. Section IV then examines the effects of surfactant addition on the mean flow and velocity fluctuations, discussing the possible micellar mechanisms underlying the observed regimes. Finally, section V summarizes the main findings and outlines directions for future work.

II. EXPERIMENTAL DEVICE

In the present study, we employ an experimental device designed to produce Taylor–Green-like forced turbulence. The apparatus, with a schematic shown in figure 1(a), consists of a water tank with four rotors which are driven by a gear system to rotate in opposing directions, thus emulating the numerical equivalent and producing turbulence. The region between the four blades, denominated bulk, is the target of measurements in the current study, and spatially resolved measurements are obtained via 2D particle tracking (briefly described in section II A). Images are captured with a high-speed camera and laser sheet combination positioned to image either the bulk in the vertical plane (figure 1(b)) or a wider region between the blades in the horizontal plane (figure 1(c)). For the imaging of the horizontal plane, a mirror is used to facilitate the positioning of the camera.

Note that between the two configurations of figures 1(b) and (c), the position of the laser remains fixed, while the lens system is rotated to change the orientation of the laser sheet. Because the proximity of the blades limits viewing area, the camera is angled slightly so that the actual measurement area sits mostly in the region between two blades as opposed to the center of the tank. Figure 2 shows the actual size and location of the visible region as the square with the velocity vectors. This configuration then allows us to measure the flow structures which develop in the region between two adjacent blades which drive the flow towards the bulk region.

As shown in the schematics of figures 1 and 2, the rotors themselves are shaped as blades with rectangular cross-section to achieve the inertial driving of the turbulence. Additionally, the rotation direction is such that

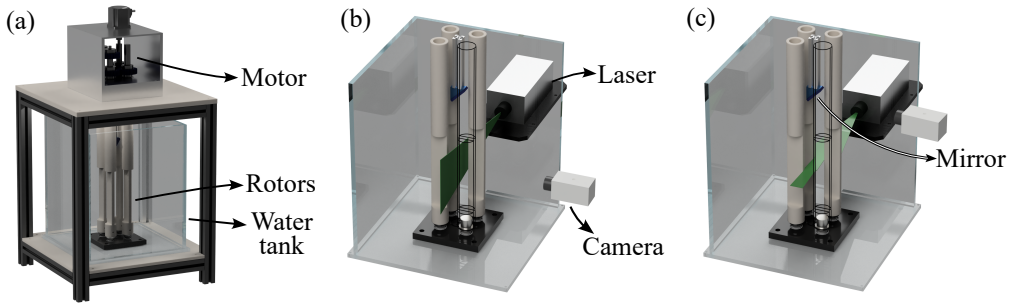


FIG. 1. Schematic (a) of the experimental device and zoomed-in views showing the laser sheet and camera orientations for the imaging of (b) the bulk and (c) the vicinity of the blades in the horizontal plane.

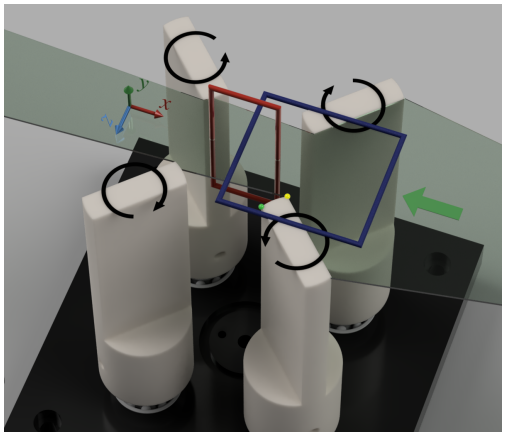


FIG. 2. Schematic of the viewing areas for imaging of the bulk (red frame) and vicinity of the blades (blue frame). The laser sheet is shown in the latter configuration, with the green arrow representing laser direction. Black circular arrows indicate direction of rotation, and green and yellow dots indicate denomination of leading and trailing edge of the blades respectively.

mean stretching of the flow happens in the out-of-plane component of the camera image (when imaging the bulk region in the vertical plane), and is accompanied by the formation of secondary vortices as shown in previous numerical studies [13]. For simplicity, the carved rotors will also hereinafter be referred to as blades.

The water tank has a horizontal cross-section of 32×32 cm and a height of approximately 35 cm, allowing it to hold 30 L of tap water. The distance between the axes of adjacent blades is 60 mm, and the blade diameter is 36 mm with a thickness of 10 mm in the carved section. The laser is a Kanomax CW532-1WM continuous-wave Nd:YAG laser operating at 532 nm. It is equipped with a cylindrical plano-concave lens and a spherical plano-convex lens, producing a laser sheet approximately 1 mm thick in the measurement region. The camera is an IDT XSM-3520 Xstream high-speed camera equipped with a Tokina 50 mm, $f/2.8$ lens, stopped down to approximately $f/4.0$ for image sharpness. The resulting spatial resolutions are 0.025 mm/pixel for bulk imaging and 0.037 mm/pixel for top-down imaging with resolutions of 1000×2560 px and 1280×1400 px respectively. While not shown in the schematics, the experimental device is also encased in thermal insulation and connected to an air-processor system. By controlling the temperature of the surrounding air, we maintain the water temperature at approximately 26 °C over the course of the experiments.

A. Data measurement and reduction

As previously mentioned, processing of the obtained images to calculate the velocity fields is conducted via a particle tracking velocimetry code developed in-house. Although the details of the algorithm are out of the scope of the current document, the code is built as a collection of previously established methods. For the detection of tracer particles from the images, we employ a variation of the iterative particle reconstruction method, which is commonly employed for volumetric particle tracking as part of the shake-the-box method [14]. The tracking of detected particle positions between the video frames is carried out by a variation of the polar coordinate-system similarity method [15].

After tracks have been identified, trajectories are filtered first with a common motion estimator to remove camera vibrations, and then according to a Wiener filter [16], which is calculated from the position spectra of the particles in a collection of trajectories. This then allows us to extract the noise floor of position estimation and determine the cutoff frequency of the filter for each of the cases. After the position of particles has been filtered, velocities are then obtained via second-order finite differencing. While two-point statistics such as the velocity structure functions are obtained directly from the Lagrangian data through a spatial binning approach, estimation of mean flow and root-mean-square (RMS) of the velocity fluctuations across the measurement domain is conducted after velocities have been interpolated into a regular grid through adaptive Gaussian interpolation [17].

Confidence intervals for mean flow and velocity fluctuations are estimated using the seasonal block bootstrap method [18], with data blocks aligned with the forcing period (half a revolution of the symmetric blades) to preserve the phase-locked structure of the results. A block length of 20 cycles, or 10 full rotations of the rotors, was selected through a sensitivity sweep study, and confidence intervals are computed using the percentile method. When presented, error bars in the results represent the 95% confidence interval.

B. Surfactant solutions

The surfactant used in the current study was the cationic surfactant cetrimonium chloride (CTAC), with the addition of sodium salicylate at a molar ratio of 4:1 to provide counterions. The solution is prepared by mixing the adequate amounts of surfactant and salt with tap water and then agitating in a magnetic stirrer until complete dissolution, after which the resulting solution is added to the experimental device. For simplicity, the surfactant and salt solution is hereinafter referred to simply as the CTAC solution.

For the analysis of the effect of solution concentration on the results, three CTAC solutions of concentrations of 33 ppm, 50 ppm, and 100 ppm were prepared. As previously mentioned, rheological measurements of such dilute solutions have historically yielded diverging results [10, 11], meaning that there is no precise measurement of properties such as the relaxation time of the micellar structures under these conditions. Additionally, the same studies indicated that the relevant properties do not scale monotonically with solution concentration, which makes estimation more difficult [10, 11].

In spite of these discrepancies, it can be expected from the aforementioned studies that the relaxation time of the solution, τ_s, should be of the order of 10^{-1} seconds, and the changes with concentration are small enough that this should remain valid across all considered cases. Still, since τ_s cannot be precisely known beforehand, or independently controlled over a wide range, a sweep over the Weissenberg number Wi is instead achieved by changing the time scale of the generated turbulence. For each of the prepared solutions, the rotation rate of the blades Ω, measured in Hz, is changed from 2^{-3} to 2^4 Hz in powers of two; for the 100 ppm solution specifically, two additional, lower rotation rates of 2^{-4} and 2^{-5} Hz are also tested, in order to further resolve the behavior of the flow in the regime of very low Wi. Defining the Reynolds number based on blade diameter D, blade tip speed $U_{\text{tip}} = \pi\Omega D$ and kinematic viscosity ν as $Re_D = U_{\text{tip}}D/\nu = \pi\Omega D^2/\nu$, the chosen rotation rates result in $Re_D \approx 151$ for the slowest case and $Re_D \approx 7.7 \times 10^4$ for the fastest. Although it is debatable whether the flow generated at the slower rotation rates can be termed turbulent, it

remains chaotic and unsteady, so such cases are still useful for determining the effect of surfactant addition on this type of unsteady, low-Reynolds-number flow.

III. TURBULENCE CHARACTERIZATION

Considering the nature of the current experimental apparatus, we characterize the turbulence generated in the Newtonian flow as a function of blade rotation rate. Although the fluctuations are anisotropic, their statistics are approximately homogeneous across the 24.6×63.0 mm bulk measurement region surrounding the stagnation point of the mean flow. We therefore calculate temporal statistics at each position and average them over this region. We denote the horizontal and vertical coordinates by x and y, respectively, with $\mathbf{x} = (x, y)$, and the corresponding instantaneous velocity components by $u(\mathbf{x}, t)$ and $v(\mathbf{x}, t)$. Their temporal means are $U(\mathbf{x}) = \langle u(\mathbf{x}, t)\rangle$ and $V(\mathbf{x}) = \langle v(\mathbf{x}, t)\rangle$. For a generic component, we write $u_x \equiv u$ and $u_y \equiv v$, with $i \in \{x, y\}$. Here and in the following sections, a prime denotes a fluctuation about the temporal mean, $u_i'(\mathbf{x}, t) = u_i(\mathbf{x}, t) - \langle u_i(\mathbf{x}, t)\rangle$, angle brackets $\langle\cdot\rangle$ denote a temporal average at fixed position, and an overbar denotes a spatial average over the stated measurement region (the bulk region here). Thus, the spatially averaged RMS of component i is

$$\overline{u_{i,\mathrm{rms}}} = \overline{\sqrt{\langle u_i'^2\rangle}}.$$

To keep the figures compact, we write $\overline{u_i'} \equiv \overline{u_{i,\mathrm{rms}}}$ for this spatially averaged RMS; $u_i'(\mathbf{x}, t)$ continues to denote the instantaneous fluctuation. Figure 3 shows $\overline{u_i'}$ normalized by tip speed as a function of the blade Reynolds number Re_D, which is proportional to the rotation rate for fixed blade diameter D and kinematic viscosity ν. Both normalized components remain approximately constant over the measured range, indicating that their spatially averaged RMS values scale approximately linearly with blade speed, $\overline{u_i'} \propto \Omega D$. This is consistent with the blade tip speed, $U_{\mathrm{tip}} = \pi\Omega D$, setting the velocity scale of the large, forced eddies. The horizontal component remains larger than the vertical component, consistent with the primary forcing direction of the blades.

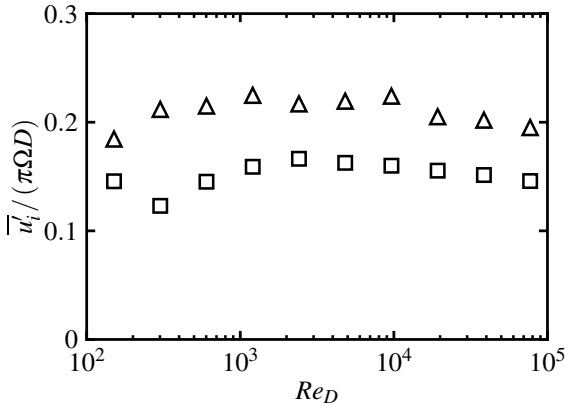


FIG. 3. Spatially averaged root-mean-square velocity fluctuations in the bulk, normalized by $\pi\Omega D$, versus the blade Reynolds number Re_D. Triangles denote the horizontal component (u); squares denote the vertical component (v).

Although direct measurement of the dissipation range is not feasible under current experimental conditions, the turbulent dissipation can be estimated from the longitudinal third-order velocity structure function. With $\mathbf{e}_i$ the unit vector along the direction of component u_i, we define

$$D_{LLL}^{(i)}(r) = \overline{\left\langle \left[u_i'(\mathbf{x} + r\mathbf{e}_i, t) - u_i'(\mathbf{x}, t)\right]^3 \right\rangle}, \tag{2}$$

where r is the separation along $\mathbf{e}_i$ and the overbar averages over positions for which both velocities are measured. In the inertial range, Kolmogorov's four-fifths law gives $D^{(i)}_{LLL}(r) \simeq -4\varepsilon r/5$, so the compensated quantity $-5D^{(i)}_{LLL}(r)/(4r)$ provides an estimate of the dissipation rate ε. Although the large scales of the generated turbulence are anisotropic, the results of figure 4(a) indicate that the dissipation estimates are approximately independent of the measurement direction. This is consistent with Kolmogorov's local isotropy hypothesis, according to which the small, dissipative scales of a turbulent flow are expected to recover statistical isotropy even when the large, energy-containing scales are not, and lends support to the validity of the obtained estimate. In all the plots, the increase in the compensated structure function at the smallest separations is caused by the finite depth-of-correlation and the nature of planar imaging, as well as the fact that no interpolation of the instantaneous velocity fields is used for the calculation of these statistics.

For the cases of high rotation rate, the appearance of a horizontal region indicates that an inertial range is well-defined, and the dissipation rate can be taken as the average across this region. The fact that the dimensionless dissipation estimates shown in figure 4(b) remain approximately constant at higher Reynolds numbers also supports the scaling of the estimate. For the lower rotation rates, the Reynolds number is too low for the appearance of a defined inertial range, and the extraction of a representative value of dissipation is more difficult. For consistency between the cases, we average the compensated third-order structure function over separations from $r = 3$ to 20 mm to estimate the dissipation rate. We denote the estimate from direction i by ε_i; these estimates are shown by the dotted lines in figure 4(a).

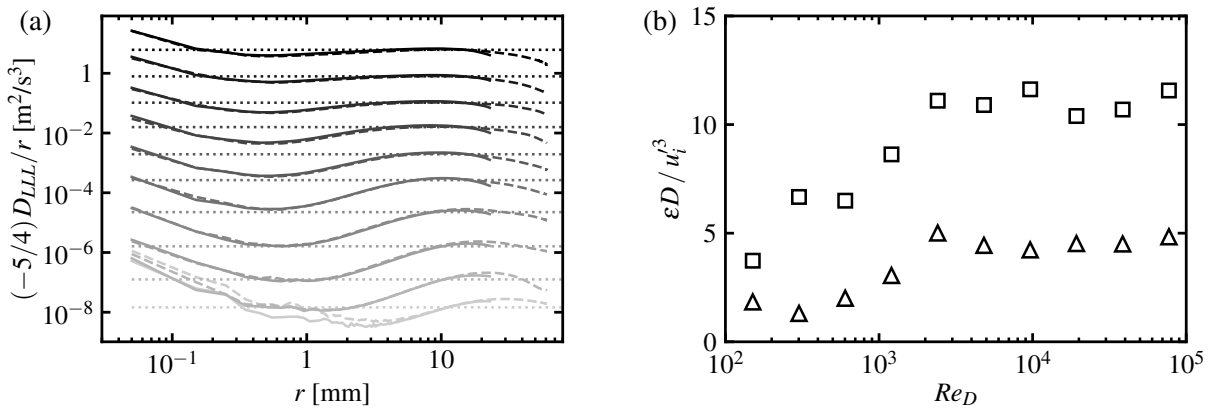


FIG. 4. (a) Compensated longitudinal third-order structure function, $-5D^{(i)}_{LLL}(r)/(4r)$, versus separation r for the Newtonian cases. Solid and dashed curves correspond to the two measured directions; darker curves indicate higher blade rotation rates. Dotted horizontal lines show the dissipation estimates obtained by averaging over $3 \le r \le 20$ mm. (b) Dimensionless dissipation estimates, $\varepsilon_i D/\overline{u_i'}^3$, versus blade Reynolds number Re_D. Triangles denote the horizontal direction ($i = x$); squares denote the vertical direction ($i = y$).

Combined with the previous scaling argument for the RMS velocity, this measurement of the dissipation rate allows us to estimate its dependence on rotation rate. In fully developed turbulence, the dissipation rate becomes independent of viscosity and is instead set by the rate at which energy is supplied to the large scales, $\varepsilon \sim \overline{u_{i,\mathrm{rms}}}^3/L$, with L a fixed characteristic length of the energy-containing eddies (again set by the geometry of the apparatus). Combined with $\overline{u_{i,\mathrm{rms}}} \sim \Omega D$, this yields the scaling $\varepsilon \sim \Omega^3$. From these results, we can then obtain the approximate scalings of the Kolmogorov length scale and time scale for each of the cases,

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4} \sim \Omega^{-3/4}, \qquad \tau_\eta = \left(\frac{\nu}{\varepsilon}\right)^{1/2} \sim \Omega^{-3/2}, \tag{3}$$

as shown in figure 5, where the measured power-law trends are consistent with the scaling above. In this case, the quantities are shown in dimensional units as they become useful comparisons for further discussions

involving the various time and length scales of the experiment. For example, the estimated Kolmogorov length scale provides us with an additional clue as to why it is not possible to directly measure the dissipation range, considering that these scales become smaller than the depth-of-correlation of the current imaging system for most rotation rates. Additionally, figure 5(b) also presents the range of Weissenberg numbers obtained for each rotation rate when considering a relaxation time of $\tau_s = 0.1$ s as per the discussion of section II B. The wide range, from $Wi = O(10^{-2})$ to $Wi = O(10^{2})$, indicates that the transition between elastically dominated and inertially dominated flow should be observed in all solutions regardless of the uncertainty in the determination of their precise relaxation time. When relevant, the results plotted in section IV include the values of Wi according to this estimate as a reference.

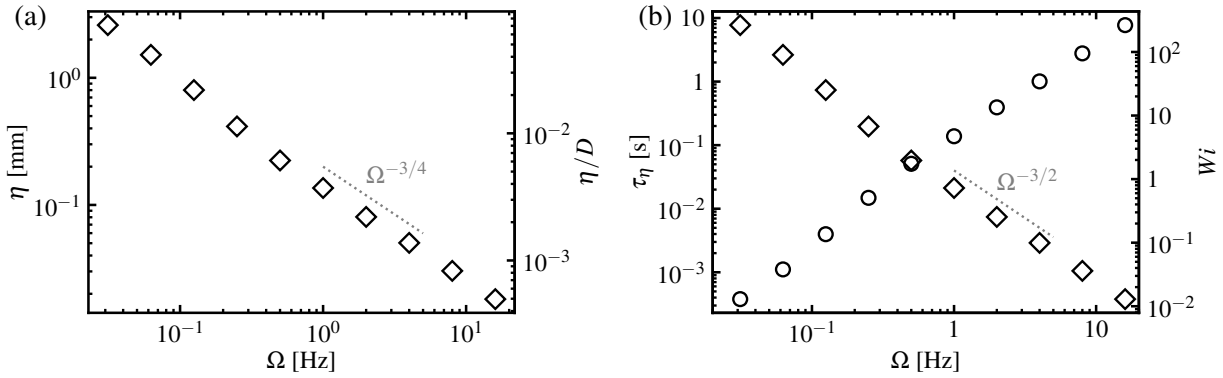


FIG. 5. (a) Kolmogorov length scale η versus blade rotation rate Ω; diamonds denote the estimates and the dotted line indicates the $\Omega^{-3/4}$ scaling. (b) Kolmogorov time scale τ_η (diamonds, left axis) and estimated Weissenberg number Wi (circles, right axis) versus Ω. The dotted line indicates the $\Omega^{-3/2}$ scaling; Wi is calculated using $\tau_s = 0.1$ s.

IV. RESULTS

The analysis of turbulence modulation by surfactant addition begins with the turbulent intensity at the bulk region, which is described here by the spatially averaged RMS of the velocity fluctuations. Since we conduct planar measurements, this magnitude is calculated from the two measured components, by defining $q'_{2D}(\mathbf{x}) = \sqrt{\langle u'^2 \rangle + \langle v'^2 \rangle}$, which is hereinafter referred to as q' for simplicity. This local RMS is then spatially averaged across the measurement region to obtain $\overline{q'} = \overline{q'_{2D}(\mathbf{x})}$, since these statistics are approximately homogeneous in this area as previously discussed. Note that although the value of q' does not contain the out-of-plane component of the turbulent kinetic energy, its magnitude can be expected to be similar to the currently measured horizontal component of the velocity fluctuations. With this in mind, the results of $\overline{q'}$ are interpreted as an indication of the trends of turbulent kinetic energy in the bulk region. Aiming to observe the modulation of such a quantity, figure 6 then presents the ratio of the spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases.

According to the time scale matching picture of turbulence modulation via the addition of surfactants and polymers, eddies with a turnover time shorter than the relaxation time of the viscoelastic solutions should be attenuated, and the turbulence in the regime where $Wi < 1$ should remain approximately unchanged. The results shown in figure 6, however, display a more complex picture. It can be seen that although turbulent energy is indeed attenuated once the Weissenberg number crosses the frontier of unity, for all cases, the turbulence also appears to be intensified for the 50 ppm and 100 ppm solutions at low rotation rates. Furthermore, once the rotation rate is increased past a certain threshold, the attenuation is lost and the solutions appear to return to the Newtonian behavior.

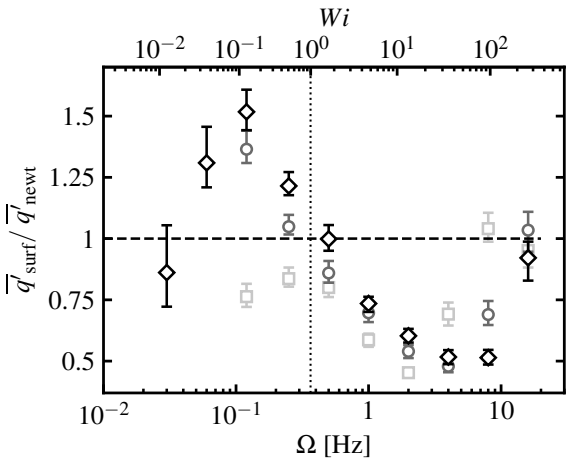


FIG. 6. Ratio of spatially averaged velocity fluctuation RMS $\overline{q'}$ between the surfactant and Newtonian cases. Vertical dashed line represents the rotation rate for which $Wi \approx 1$ considering a relaxation time of $\tau_s = 0.1$ s. Darker markers indicate higher solution concentration, with the squares denoting 33 ppm, circles denoting 50 ppm and diamonds denoting 100 ppm solution.

To understand these phenomena, we first examine the effect of surfactant addition on mean flow intensity around the stagnation point. Considering the symmetric nature of the experimental apparatus, as well as the forcing with respect to the right and left regions of the measurement domain, we can enforce this symmetry on the mean flow to improve convergence of the results. This is done by defining a mirror coordinate system as $\tilde{\mathbf{x}} \equiv (2x_c - x,\ y)$, with $x = x_c$ corresponding to the symmetry axis, or stagnation point. This then allows us to write a symmetry-enforced mean flow field component, defined as:

$$U^{\dagger}(\mathbf{x}) = \frac{1}{2}\left[U(\mathbf{x}) - U(\tilde{\mathbf{x}})\right]. \tag{4}$$

We can then compute the spatial average of this quantity to then obtain a representative intensity of the mean flow. Since the field is antisymmetric, however, this spatial average is computed only on the left-hand side of the stagnation point (such that $x < x_c$), resulting in the quantity $\overline{U}^{\dagger}_{L}$. Note that this approach is chosen (in opposition to taking the spatial average of the velocity magnitude) because it allows us to preserve the flow direction, and thus take into account flow reversal as well should it happen. Moreover, since the defined quantity has the significance of "spatially averaged mean flow" in the bulk region, we write $\overline{U} \equiv \overline{U}^{\dagger}_{L}$ for visual clarity, with the symmetry enforcement and left-half averaging retained in this shorthand. With this in mind, figure 7 presents the ratio of mean flow intensity between the surfactant and Newtonian cases.

From the plot of figure 7, it can be seen that for rotation rates of approximately 1 Hz or above, the addition of surfactant has almost no effect on the mean flow, irrespective of the concentration. It can be stated that such would be the expected behavior for turbulence which is inertially driven, since the time scale matching picture of turbulence modulations states that the role of surfactant additives is in the attenuation of the small scales. Since the driving rotors act on the large scales, which even at the slowest rotation rate are larger than the Kolmogorov length scale by a factor of at least one decade (ref. fig. 5), it can be expected that such large scales should remain mostly unaffected. Nevertheless, the results from figure 7 indicate that reducing the rotation rate leads to an increase of up to 4 times in the mean flow intensity for the regime where $\Omega \lesssim 1$ Hz. Additionally, this behavior appears to be independent of the frontier of $Wi \approx 1$, with the peak in mean flow intensification occurring well below the value of unity of the Weissenberg number, after which the amplification effect appears to cease once the rotation rate is further decreased.

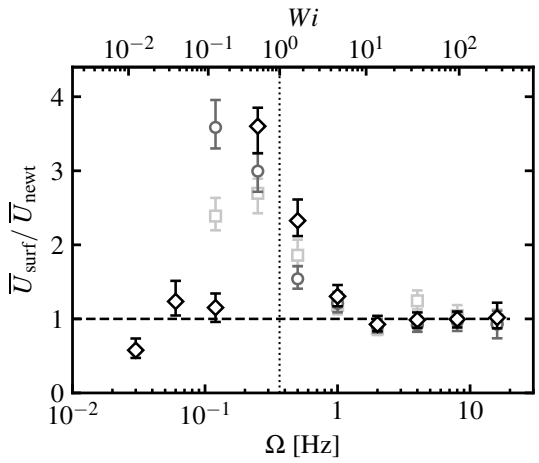


FIG. 7. Ratio of the signed, spatially averaged horizontal mean flow component $\overline{U}$ between the surfactant and Newtonian cases. Vertical dashed line represents $Wi \approx 1$. Darker markers indicate higher solution concentration, with the squares denoting 33 ppm, circles denoting 50 ppm and diamonds denoting 100 ppm solution.

To investigate the cause behind this increase in the mean flow for the lower rotation rates, figure 8 presents the value of the same spatially averaged horizontal component of the mean velocity, but as a function of the phase of the blade (ϕ). We define the local phase-conditioned means as $U_\phi(\mathbf{x}, \phi) = \langle u(\mathbf{x}, t)\rangle_\phi$ and $V_\phi(\mathbf{x}, \phi) = \langle v(\mathbf{x}, t)\rangle_\phi$, where $\langle\cdot\rangle_\phi$ denotes an average over samples at the same blade phase across successive cycles. Similarly to the previously defined quantities, we introduce $U^\dagger_\phi(\mathbf{x}, \phi) = \frac{1}{2}\left[U_\phi(\mathbf{x}, \phi) - U_\phi(\tilde{\mathbf{x}}, \phi)\right]$ corresponding to the symmetry-enforced version of the phase-locked component of the mean flow U_ϕ. Again, for visual clarity, this quantity is hereinafter simply referred to as $\overline{U_\phi}$ after the spatial averaging over the left side of the observed domain is taken.

From these plots, it is possible to see that the overall dependence of the mean flow on the phase of the blade gets amplified as either the concentration of the solution is increased or the rotation rate is decreased, possibly leading to a reversal of flow direction as well. For the 50 and 100 ppm solutions, decreasing the rotation rate produces asymmetric oscillations, with a strong increase in mean flow intensity for most of the cycle and only a small decrease over a narrow phase interval. This behavior reaches an extreme around the rotation rate of 2^{-2} or 2^{-3} Hz (depending on solution concentration) where the intensification of the mean flow reaches a peak. To aid the analysis, we define the oscillation amplitude as half the peak-to-peak range, $A_{U_\phi} = \frac{1}{2}\left[\max_\phi \overline{U_\phi}(\phi) - \min_\phi \overline{U_\phi}(\phi)\right]$. Figure 9 presents this amplitude normalized by the blade tip speed $\pi\Omega D$ for each of the cases.

The amplitude curve for each of the solutions indicates that while the Newtonian flow displays an approximately constant normalized amplitude, the surfactant solutions show a peak in normalized amplitude at the frequency where the oscillations are most pronounced. While data for the slowest rotation rates of the 50 ppm solution is not available, analysis of the curve shapes from figure 8 indicates that the oscillations grow in a similar manner between the cases, and that the oscillation peak of the 50 ppm solution should happen at a lower frequency than the 100 ppm counterpart.

For the 100 ppm solution, the two lowest rotation rates produce more symmetric oscillations and a pronounced interval of flow reversal. Even though the rotation of the blades is always in the sense of pushing flow towards the central region (as seen from the point of view of the camera), an opposite movement is observed, with the flow going from the center towards the lateral edges of the domain. On the opposite end of the spectrum, it can be seen from both plots that past a certain rotation rate (approximately 4 Hz in the current experiment), the oscillations cease, and the three solutions recover the Newtonian behavior, similarly to what is seen in the plot of figure 7.

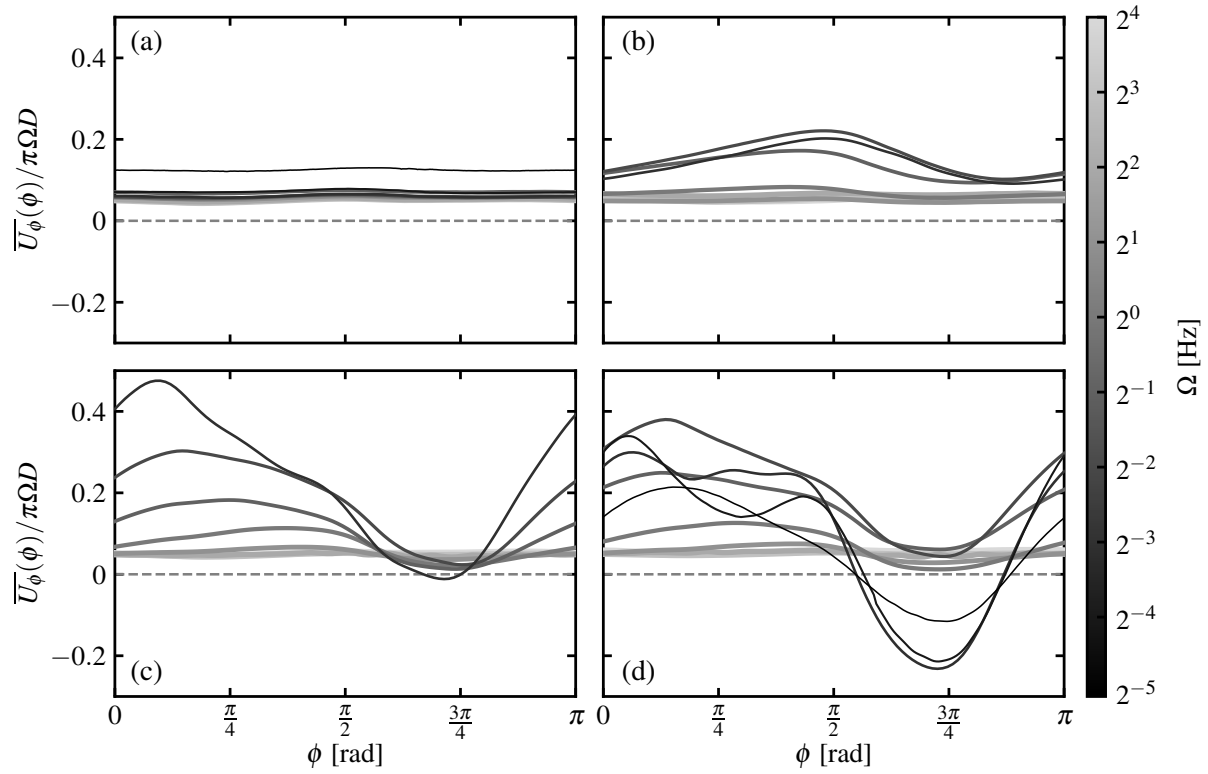


FIG. 8. Signed horizontal mean flow component $\overline{U_\phi}$, symmetry-enforced and spatially averaged over the left half of the measurement region, as a function of blade phase for (a) Newtonian, (b) 33 ppm, (c) 50 ppm and (d) 100 ppm solutions. Positive value refers to inward motion (towards the stagnation point), while a negative value indicates flow reversal.

The behavior observed in figures 8 and 9 is indicative of a resonance phenomenon between the flow forcing and large structures which have a relaxation time much longer than the typical relaxation time of the micelles, which, as previously described, is of the order of 0.1 s. Considering that the current resonant peaks appear at a rotation rate of approximately 0.1 Hz (and therefore a forcing frequency of 0.2 Hz due to the geometry of the blades), it can be expected that the structures at play in this case also have a relaxation time of the order of 5 s. Moreover, for them to interact with the blades and produce the observed scale of velocity oscillation, the spatial scale of these structures can also be expected to be of the order of 1 cm, the same as the spacing between the blades.

In order to analyze the movement and behavior of such flow structures in more detail, we can rely on the flow fields measured in the horizontal plane and in the vicinity of the blades according to the schematics of figure 1(c) and 2. Figure 10 displays the evolution of the flow field over the course of half a rotation period divided into four snapshots, for the Newtonian and 50 ppm cases at the rotation rate of 0.125 Hz, divided into the top and bottom rows respectively. In both cases, the evolution is displayed from left to right, with the blades initially in the position of largest constriction of the flow. From there, each consecutive snapshot is delayed by a rotation of 45°. For the Newtonian flow case, in the top row, no particularly noteworthy flow structures are visible, with the highest flow velocity seen in the vicinity of the blades (particularly behind the trailing edge), and a general attenuation of the velocity magnitude towards the bulk region. For the surfactant case in the bottom row, however, more complex flow patterns are observed.

Analyzing the flow structures of the surfactant case, in the bottom row of figure 10, we start at the first snapshot where the blade edges are at their closest approach. In this orientation the motion of the blades is accompanied by broader (when compared to the Newtonian flow) higher velocity regions which extend from the leading edge towards the trailing edge, behind the movement of the blades. In the second snapshot, as the

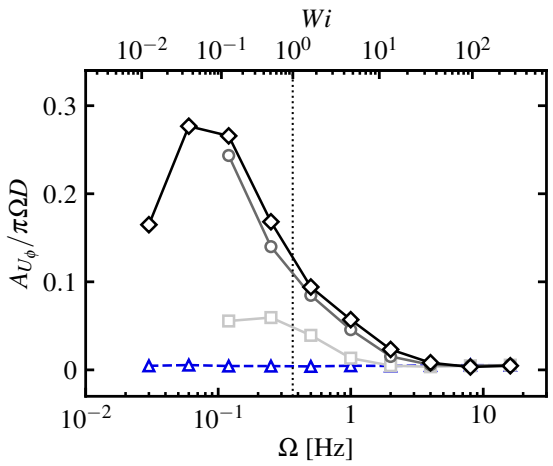


FIG. 9. Oscillation amplitude of the spatially averaged, phase-locked horizontal component of the mean flow, defined as half the peak-to-peak range and normalized by blade tip speed, $A_{U_\phi}/(\pi\Omega D)$, as a function of rotation rate. Grayscale curves denote surfactant solutions with darker markers indicating higher solution concentration. Squares denote 33 ppm, circles denote 50 ppm and diamonds denote 100 ppm solution. The blue line with triangle markers denotes Newtonian flow.

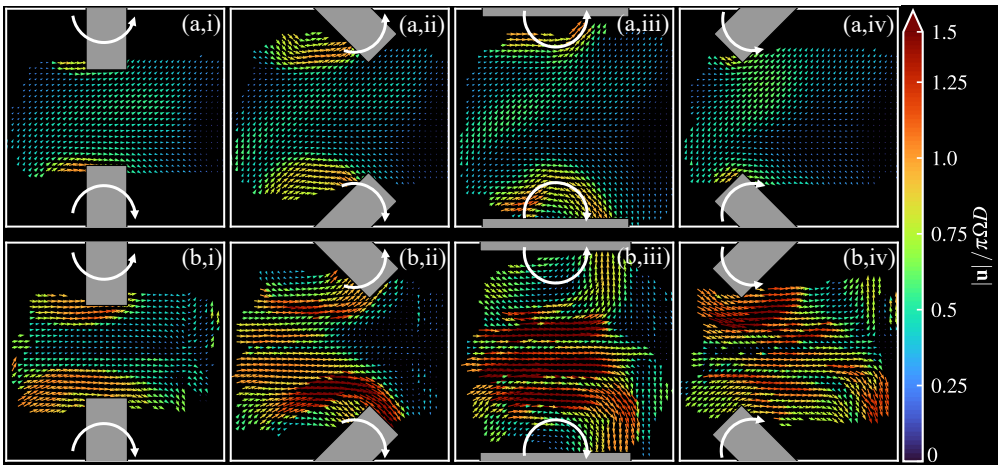


FIG. 10. Top-down visualization of the flow field from Newtonian (a) and 50 ppm (b) cases at rotation rate of 0.125 Hz. For the top and bottom rows, plots (i) through (iv) represent four consecutive snapshots separated by a phase difference of $\pi/4$.

two blades begin to separate, these two faster moving regions transform into a more developed shear-layer structure, with an increase in the flow velocity of the layers in direct proximity to the blade and initial formation of secondary high velocity structures in the space between the blades.

At the third snapshot, the layered structures continue to develop, and the velocity magnitude in the region between the blades also reaches its maximum value. Note that the velocity of these streaks, or jet-like structures, surpasses the tip velocity of the blades, and is strongest at the position of largest separation of the blades. It can be hypothesized that as the leading and trailing edges of the blades drag large micellar networks, these are initially compressed as the blades reach close proximity, and then relax, leading to a pronounced acceleration of the flow in this condition. As these "jets" then impinge into the bulk, this region itself receives lateral compression which relaxes once the jets disappear, leading to the reduction in mean flow intensity and possibly flow reversal in this region.

In the fourth snapshot of figure 10 we can see that the jet-like structures appear to break down, or more simply lose strength, in the measured region, while an increase of velocity in the top and bottom right portions of the domain can be seen, likely signaling that these structures are now being pulled apart and downwards or upwards for each of the blades. As we return to the first snapshot, the constriction of the flow by the two blades rapidly leads to the attenuation of the high velocity regions between the blades, which is also consistent with the rapid drop in velocity which is seen in figure 8.

The streak-like structures shown in figure 10 show resemblance to the elongated structures observed in drag-reduction studies in channel flow using solutions of similar concentrations [19]. In the aforementioned study, the authors argue that the formation of shear-induced structures (SIS) is a key part of the drag-reducing capability of surfactant, and the spatial measurements of flow velocity in the wall-parallel planes of the drag reducing solutions show similar elongated structures in the vicinity of the walls. In our current experiment, there exists a region of mean shear between the blades due to the velocity gradient up to the bulk region, where the mean flow is stagnant. It can be hypothesized that this shear, coupled with the relatively low turbulent intensity of the slower rotation rate cases, leads to favorable conditions for the formation of SIS, which we see as the jet-like structures of figure 10.

The decrease in oscillation amplitude at still lower rotation rates is consistent with a mismatch between the forcing period and the response time of the large structures, supporting the idea of resonance.

Having established the importance of blade phase on the state of the mean flow in the current experiment, we can now turn our analysis to the evolution of the velocity fluctuations. For this, we apply the triple decomposition technique, which essentially consists of decomposing the flow field into its mean, periodic, and chaotic components [20], in order to remove the contribution of the periodic forcing which would otherwise be included as part of the velocity fluctuations. Essentially, we subtract the local phase-conditioned mean velocity field, evaluated at the instantaneous blade phase $\phi(t)$, from the instantaneous flow field in order to obtain the chaotic component of the velocity fluctuations, with magnitude simply defined as (5) to keep consistency with previously defined quantities. Here, the prime in q'_ϕ denotes residual fluctuations about the phase-conditioned mean, rather than fluctuations about the temporal mean. The ratio of this spatially averaged, phase-corrected RMS $\overline{q'_\phi}$ between the surfactant and Newtonian cases is then presented in figure 11.

$$q'_\phi(\mathbf{x}) = \sqrt{\left\langle \left(u(\mathbf{x},t) - U_\phi(\mathbf{x},\phi(t))\right)^2 \right\rangle + \left\langle \left(v(\mathbf{x},t) - V_\phi(\mathbf{x},\phi(t))\right)^2 \right\rangle} \tag{5}$$

When contrasted with the results obtained without the triple decomposition (cf. fig. 6), the results of figure 11 show a less pronounced increase in the turbulent energy for low rotation rates, and the picture becomes closer to what would be expected according to the time matching argument with a single relaxation time. Most notably, we see attenuation of the turbulent energy once the Weissenberg number crosses the frontier of unity (approximately), with an attenuation which is proportional to the rotation rate in the vicinity of this frontier. In this region, the turnover time of the smallest eddies becomes shorter than the relaxation time of the string-like micellar structures, which are then extended and begin to extract energy from these small scales in the process. As the Reynolds number is increased, more turbulent scales begin to interact with the micelles and the attenuation becomes more pronounced. The onset of attenuation near the estimated $Wi = 1$ is a good indication that the relaxation time of the solutions is of the order of 0.1 seconds. The presence of other effects, however, such as the resonance with large-scale structures, complicates any comparison of the relaxation time between the three solutions.

Past the attenuation region, as the Reynolds number is further increased, we see another change in the trend of turbulent intensity: the attenuation effect is diminished, and the turbulent intensity progressively returns to the value of the Newtonian fluid once again. This behavior is analogous to what has been observed in studies of drag reduction in channel flow [19, 21], where a critical Reynolds number exists, past which the drag reduction effectiveness sharply declines and the flow returns to the Newtonian state. In such studies, the authors argue that while the presence of some shear is essential for the formation of shear-induced structures

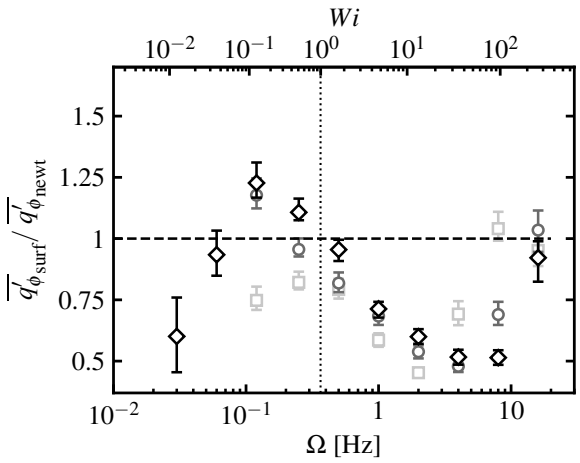


FIG. 11. Ratio of spatially averaged, phase-corrected velocity fluctuation RMS $\overline{q'_\phi}$ between the surfactant and Newtonian cases. Darker markers indicate higher solution concentration, with the squares denoting 33 ppm, circles denoting 50 ppm and diamonds denoting 100 ppm solution.

which play a key role in the drag reduction, excessive shear leads to the breakup of such structures and inhibits their formation, which undermines the drag reduction effectiveness.

We propose that this process happens in two main steps, linked to the observed regimes of turbulence intensification and attenuation. Initially, for very slow rotation rates, the mean shear favors the formation of very large SIS, which then interact with the forcing frequency, leading to the previously discussed resonance phenomenon. As the rotation rate is increased, these structures break down, or their formation is simply not favored, but the individual string-like micellar structures remain intact. In the second regime, the turnover time of the smallest eddies becomes shorter than the relaxation time of these small micellar structures, and the attenuation of turbulence begins. However, as the rotation rate is further increased, and we approach the threshold of $Wi = O(10^2)$ in the bulk, the shear rate becomes large enough that these smaller micellar structures break down. The resulting smaller micelles should then have much faster relaxation times, and the turbulence attenuation mostly ceases.

Here it is also important to note that we believe the geometry of the blades themselves might play an important role in the rotation rate at which the breakdown happens in both regimes. The rectangular cross-section of the blades, while useful in the sense that it allows us to isolate the inertial component of the flow forcing, also generates regions of very high shear rate in the vicinity of the leading and trailing edges. This in turn means that the Weissenberg number in this region is considerably higher than that of the bulk, and the breakdown of the micellar structures might be concentrated in this region, especially in the case of the second regime at very high rotation rates.

Another possible hypothesis for the behavior observed in the loss of turbulence attenuation is that before the breakdown process begins, the string-like micellar structures are saturated (at maximum extension) and are therefore not able to absorb more energy, even though the energy input rate is being increased with the increase in rotation rate. This hypothesis would then be consistent with the observation that the rotation rate at which the turbulent energy ratio returns to the vicinity of unity is proportional to the concentration of the solutions, with this process happening first for the 33 ppm solution and being most delayed for the 100 ppm solution. Nevertheless, the unclear rheological characteristics of the solutions make it difficult to precisely pin down the mechanisms at play here.

A final observation is that the alignment of large micellar structure breakdown with a Weissenberg number close to unity is a coincidence brought forth by the characteristics of the current experiment. It is still unclear whether the resonant frequency of these very large scale structures is determined mostly by

the geometrical constraints of the experimental device, or simply by the solution itself (most likely), and also how the ratio between the two observed characteristic times changes depending on concentration and solution chemistry.

V. CONCLUSIONS

In this work, we conducted an experimental investigation of the turbulence modulation caused by the addition of a dilute cationic surfactant (CTAC/NaSal) to a Taylor–Green-like forced flow, using blade-shaped rotors to isolate inertial forcing over a wide range of Reynolds numbers and Weissenberg numbers. By combining measurements of the mean flow, phase-resolved analysis of the velocity field, and triple decomposition of the turbulent fluctuations, we were able to show that the turbulence modulation by surfactant addition departs from the explanation of time scale matching between a single relaxation time of the solutions and the time scales of the flow.

Instead, our results indicate that the response of the flow to the surfactant additive can be divided into two distinct and possibly overlapping regimes with unique time scales, as a function of forcing period. In the first regime, which occurs at the lowest rotation rates ($\Omega \lesssim 1$ Hz), the surfactant additive surprisingly leads to an intensification of both the raw velocity fluctuations and the mean flow. We were able to show that this intensification is tied to a resonance-like interaction between the periodic forcing of the blades and large, elastic micellar structures with a much longer relaxation time (of the order of seconds) than that inferred for the small-scale turbulent fluctuations. This resonance also leads to a strong dependence of mean flow intensity on the agitator phase, and possibly even reversal of flow direction. A direct visualization of the flow field in the vicinity of the blades revealed the mechanism behind this behavior: the periodic compression and relaxation of these large-scale structures by the movement of the blades generate streak- or jet-like structures which accelerate well beyond the tip speed of the agitators, and directly drive the observed intensification of the mean flow in the bulk region.

Once the periodic nature of the velocity fluctuations is properly accounted for via triple decomposition, the resulting phase-corrected velocity fluctuations reveal a picture which is much closer to the time scale matching argument. In the second regime, as the Weissenberg number crosses the value of unity, turbulent energy decays as the turnover time of the smallest eddies becomes shorter than the relaxation time of the string-like micelles, which are then stretched and extract energy from the turbulent cascade. As the Reynolds number increases, so does the attenuation of turbulent energy, until each of the solutions reaches a threshold beyond which the attenuation ceases and the velocity fluctuations recover the behavior of Newtonian flow, characterizing the end of the second regime. In this final stage, we attribute the loss of turbulence attenuation to the shear-induced breakdown of the micellar network, a behavior analogous to the critical-Reynolds-number effects which have been reported in channel-flow drag reduction studies.

Taken together, our results suggest that the modulation of turbulence in bulk, inertially driven flow with dilute surfactant solutions cannot be reduced to a single attenuation mechanism governed by a single relaxation time. Rather, at least two distinct groups of viscoelastic structures appear to be relevant, consisting of the large-scale structures which can resonate with the periodic forcing, and the small-scale, string-like micelles which are responsible for the attenuation of small-scale turbulent eddies. We propose a qualitative two-step scenario which connects the effects of these two groups and leads to the resulting modulation of turbulence, but we also stress that many aspects of this picture remain speculative. In particular, whether the breakdown of the large-scale structures coincides with $Wi \approx 1$ as a consequence of the specific geometry and forcing frequency of the current apparatus, or reflects a more general property of the surfactant solution, cannot be settled with the present data. Likewise, the possibility that the loss of attenuation at high rotation rates is caused by saturation of the micellar extension, rather than by outright structural breakdown, remains an open hypothesis, motivated only indirectly by the observed dependence of this transition on solution concentration.

Future work should aim to disentangle these possibilities and some of the ambiguities in the current observations. For instance, independently varying the geometrical length scale of the experimental apparatus, as well as the geometry of the blades, in a more systematic manner should allow us to pin down whether these parameters modify the relaxation time and breakdown point of the large-scale structures in the first regime. Likewise, changing the chemistry of the solutions should allow us to verify whether there is a direct link between the relaxation time of the small structures and the larger networks, or whether these regimes represent truly independent behaviors. Moreover, considering viscously driven turbulence should help us clarify whether the mechanisms observed here are specific to inertially driven turbulence, or are a more general characteristic of surfactant solutions subject to periodic straining.

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